

ABHINAY KUMAR NALBAND

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PROFESSIONAL SUMMARY

Software Engineer and M.S. Computer Science student with 2+ years of professional experience building Java/Spring Boot backend applications, RESTful APIs, microservices, and database-driven systems, along with hands-on experience in machine learning, computer vision, NLP, and RAG-based AI applications. Skilled in Java, Python, Spring Boot, Spring Security, Hibernate/JPA, React.js, Angular, FastAPI, Flask, SQL, Docker, Azure, OpenCV, YOLO Pose, CVAT, scikit-learn, vector databases, and LLM APIs. Experienced in developing secure backend services, preparing ML datasets, training and validating models, building API-driven AI systems, and delivering production-ready software in Agile environments.

TECHNICAL SKILLS

- **Languages:** Java, Python, SQL, JavaScript, TypeScript, C++, C
- **Backend & Full Stack:** Spring Boot, Spring MVC, Spring Security, JWT, Hibernate/JPA, Maven, RESTful APIs, Microservices, Spring Cloud Netflix Eureka, FastAPI, Flask
- **Frontend:** Angular, React.js, HTML5, CSS3
- **Machine Learning & AI:** Computer Vision, Deep Learning, NLP, Object Detection, Image Classification, Keypoint Detection, Pose Estimation, RAG, Multi-Agent AI
- **ML / AI Tools:** OpenCV, YOLO Pose, CVAT, scikit-learn, NumPy, pandas, Matplotlib, Jupyter Notebook
- **Databases:** MySQL, PostgreSQL, Oracle, MariaDB
- **Cloud, DevOps & Testing:** Microsoft Azure, Docker, Jenkins, GitHub Actions, CI/CD, JUnit, Mockito, SonarLint, Swagger/OpenAPI, Postman
- **Development Tools & Practices:** Git, GitHub, Bitbucket, Jira, Agile/Scrum, Code Review, Debugging, Database Design, API Testing

WORK EXPERIENCE

Graduate Research Assistant - Texas A&M University - Aug 2025 - May 2026

- Developed computer vision workflows for fish and shrimp species detection, body measurement, and length/size/weight estimation using image and video datasets.
- Annotated fish and shrimp datasets using CVAT and prepared structured training data for YOLO Pose-based keypoint detection and pose estimation models.
- Trained and evaluated YOLO Pose and contour-based OpenCV models to detect body landmarks, support measurement extraction, and automate visual analysis workflows.
- Built reusable preprocessing pipelines for image cleaning, annotation conversion, dataset organization, train/validation splitting, model training, and result validation.
- Evaluated model outputs using detection and localization metrics such as precision, recall, F1-score, mAP, and qualitative error analysis.
- Documented experiment workflows, model results, and dataset preparation steps to support reproducible research and academic deliverables.

Graduate Assistant - Web & Technical Support, Texas A&M University - Sep 2024 - Jun 2025

- Maintained departmental web pages by implementing front-end content updates, page layout changes, navigation improvements, and usability enhancements for student- and faculty-facing pages.
- Coordinated with faculty and university stakeholders to gather website update requirements, validate requested changes, and deliver accurate technical updates within academic deadlines.
- Performed web quality checks to identify broken links, outdated content, formatting issues, accessibility gaps, and inconsistent page structures.
- Supported web troubleshooting, content deployment validation, and technical documentation for departmental academic pages.

Associate Software Engineer - DXC Technologies - Aug 2022 - Aug 2024

- Developed and deployed Java/Spring Boot microservices using Hibernate/JPA, Maven, and REST API design principles, reducing inter-service dependencies by 40% and improving backend development efficiency.
- Designed scalable backend architectures and relational database schemas for enterprise applications using SQL-based databases, improving system performance by 25%.
- Implemented backend modules for enterprise workflows by integrating service-layer logic, database operations, REST API endpoints, request validation, and error-handling mechanisms.
- Containerized Spring Boot applications using Docker and integrated service discovery with Spring Cloud Netflix Eureka, improving service scalability and reducing lookup latency by 25%.
- Improved backend code quality and reliability through SonarLint static analysis, JUnit test cases, and Mockito-based unit testing, reducing post-deployment defects by 30%.
- Collaborated in Agile/Scrum sprint cycles using Jira, Bitbucket, and code reviews to resolve defects, deliver features, and support production-ready releases.

Technical Intern - Rakuten PVT. Limited - Jan 2022 - Jun 2022

- Contributed to development, testing, and quality assurance of cloud-based application services, accelerating feature delivery by 15% through debugging, validation, and sprint-based implementation.
- Implemented and enhanced 5+ application features by analyzing requirements, updating service logic, and validating expected functionality across development cycles.
- Resolved 30+ defect and bug tickets using Jira and Bitbucket within Agile sprint cycles, maintaining a 95% task completion rate and

improving release readiness.

- Validated RESTful APIs using Postman by testing request/response behavior, status codes, payload accuracy, and service-to-service communication reliability.
- Collaborated with developers and QA teams to reproduce issues, document defects, verify fixes, and reduce recurring service errors by 10%.

Undergraduate Research Assistant - Image Classification Project - Mar 2021 - Aug 2022

- Supported a faculty-led deep learning image classification project by preparing datasets, preprocessing images, assisting model training, and analyzing experiment results.
- Organized image datasets for training and validation workflows, including cleaning, labeling support, resizing, normalization, and preprocessing.
- Assisted with model testing, result documentation, and error analysis, gaining early experience with Python-based deep learning workflows.

Web Developer - Vignan Institute of Technology and Science - May 2020 - Jan 2021

- Developed and maintained college website pages using HTML5, CSS3, JavaScript, and responsive design practices to improve layout consistency, usability, and content presentation.
- Implemented front-end UI updates, page formatting changes, navigation improvements, and content revisions based on faculty, department, and project requirements.

EDUCATION

Texas A&M University, USA

Master of Science in Computer Science

Vignan Institute of Technology and Science, India

Bachelor of Technology in Computer Science

ACADEMIC PROJECTS

Talk2Campus - API-Driven Campus Information Assistant

Repository: github.com/PushkarSikharam/Talk2Campus

Tech Stack: React.js, FastAPI, Python, REST APIs, RAG, Vector Database, LLM APIs, WhisperX, Git

- Built a full-stack RAG-based campus assistant using React.js and FastAPI to provide conversational access to distributed university information.
- Designed a retrieval-augmented generation pipeline separating frontend, API layer, retrieval module, vector database, and LLM response orchestration.
- Implemented and validated 10 REST API endpoints for query handling, retrieval, response generation, and user-facing campus information workflows.
- Integrated vector database search with LLM APIs to retrieve relevant context before generation and improve answer relevance.
- Added WhisperX-based voice support to enable speech-assisted campus queries and improve accessibility.

Emotive Chatbot - NLP Emotion Classification System

Tech Stack: React.js, Flask, Python, NLP, Sentiment140

- Built an NLP-based chatbot using Python, Flask, React.js, and Sentiment140 to classify user emotions and generate emotion-aware responses.
- Applied text preprocessing techniques including tokenization, stop-word removal, text cleaning, and feature extraction for sentiment/emotion classification.
- Trained and evaluated ML models using accuracy, precision, recall, F1-score, and confusion matrix analysis to validate classification performance.
- Developed Flask API endpoints for backend inference and integrated them with a React.js interface for real-time chatbot interaction.

Fish/Shrimp Detection and Measurement System

Tech Stack: Python, OpenCV, YOLO Pose, CVAT, NumPy, pandas, Matplotlib

- Built a computer vision pipeline for fish/shrimp detection, pose estimation, and body measurement using YOLO Pose and OpenCV-based image processing.
- Annotated image and video datasets in CVAT and prepared structured labels for model training, validation, and keypoint detection workflows.
- Applied contour detection and computer vision techniques to support fish/shrimp body structure analysis, measurement extraction, and length/size/weight estimation.

Micro-Ecommerce Application

Tech Stack: Java, Spring Boot, Spring MVC, Spring Security, JWT, Hibernate, JPA, Maven, RESTful APIs, Microservices, Spring Cloud Netflix Eureka, Angular, React.js, TypeScript, MySQL, Docker, JUnit, Mockito, Swagger/OpenAPI, Postman, Git

- Developed a secure full-stack e-commerce application using Spring Boot, Angular/React.js, MySQL, and Docker with product browsing, cart management, order tracking, checkout workflows, and user authentication.
- Implemented JWT-based authentication and authorization using Spring Security to secure REST API endpoints and manage role-based access for users and administrators.
- Designed RESTful APIs for product catalog, cart operations, order management, user authentication, and checkout workflows using Spring MVC and layered backend architecture.
- Integrated Hibernate/JPA with MySQL to manage entity mappings, repository layers, relational database operations, and transactional business logic.
- Documented and tested APIs using Swagger/OpenAPI, Postman, JUnit, and Mockito to validate service behavior and improve backend reliability.

Certifications

- Microsoft Azure Fundamentals (AZ-900)